Subject: Manufacturing Processes (23A0340IT)

UNIT - 1 : CASTING

Steps in casting, patterns & allowances, molding, cores, gating & risers, melting furnaces, solidification, defects, centrifugal / die / investment / shell casting

1. Introduction & Steps in Making a Casting

Casting is a manufacturing process in which molten metal is poured into a mould cavity having the shape of the desired part and is allowed to solidify. The solidified part is called a casting. It is one of the oldest shaping processes (about 6000 years old).

Casting = Melting + Pouring + Solidification + Removal + Finishing.

Steps involved in making a casting

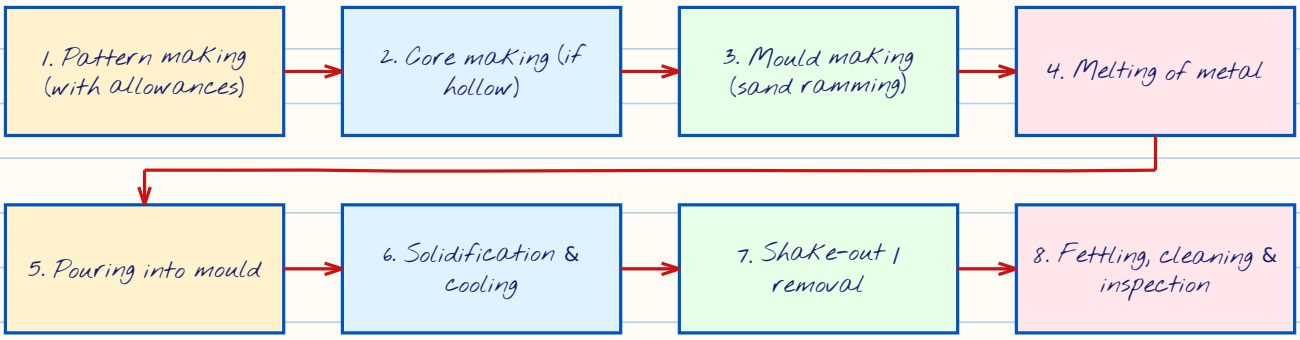


Fig 1.1 Sequence of operations in sand casting

- Pattern: replica of the part, made slightly larger (allowances) - used to make the cavity in sand.
- Mould: prepared by ramming moulding sand around the pattern in two halves - cope (upper) and drag (lower) - then pattern is withdrawn.
- Core: separate sand body placed in the cavity to form internal holes / hollow portions.
- Gating system: sprue, runner, gates and riser cutlplaced in the mould for metal flow & feeding.
- Fettling: removal of gates, risers, flash, sand; then grinding, heat-treatment & inspection.

Advantages of casting

- Very complex shapes (internal cavities, undercuts) can be made in a single piece.
- Almost any size - from a few grams (dental parts) to 100+ tonnes (ship propellers, machine beds).
- Practically any metal can be cast (ferrous & non-ferrous) - even those difficult to forge/machine.
- Low tooling cost for small quantities - economical for both one-off and mass production.
- Near-net shape: less machining & material wastage; scrap can be re-melted.
- Good vibration damping (cast iron) and isotropic properties; strong parts like engine blocks possible.

Limitations

- Dimensional accuracy & surface finish of sand casting are poor; machining often needed.
- Defects like porosity, blow holes, shrinkage are common; labour-intensive; mechanical properties are lower than wrought products.
- High-temperature metal handling - safety & pollution (fumes, silica dust).

Applications

Component	Examples
Automobile	Engine block, cylinder head, piston, manifolds, brake drums, gear-box housing
Machine tools	Lathe bed, machine bases, frames, gear blanks, pulleys, flywheels
Power & Heavy	Turbine casings, pump impellers, valves, boiler fittings
General	Pipes & fittings, manhole covers, sanitary ware, statues, art pieces, jewellery

2. Patterns & Pattern Making

A pattern is a model/replica of the object to be cast, used to form the cavity in the mould. It is normally made larger than the final casting to account for metal shrinkage and machining.

Functions of a pattern

- Forms the impression (cavity) of the casting in the mould.
- Carries core prints that locate cores; helps in establishing the parting line.
- Pattern also provides runners, gates & risers (in gated / match-plate type) and reduces defects.
- Helps in minimising overall cost - many moulds can be made from one pattern.

Types of patterns

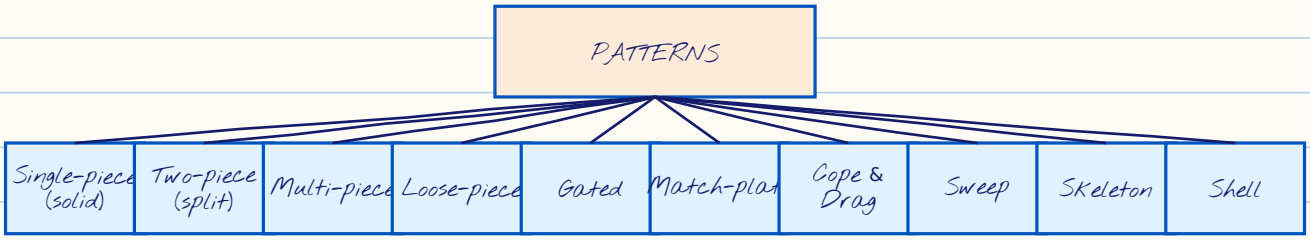


Fig 2.1 Classification of patterns

Type	Description & use
Single-piece / solid	One-piece, simplest, shape easy; parting line is flat. Used for simple shapes, low volume (e.g. flat plate, bush blank).
Split (two-piece)	Split at the parting line into 2 halves aligned by dowel pins. For moderately complex shapes - spindles, cylinders, pipes.
Multi-piece	Three or more parts (cope, cheek, drag) for very complex castings.
Loose-piece	Has projections/loose parts which are withdrawn separately after moulding - for shapes having undercuts.
Gated	Several patterns connected by gates/runners; mass production of small castings - saves moulding time.
Match-plate	Cope & drag patterns mounted on opposite sides of a metal plate, with gates; used with moulding machines; high accuracy & large quantity.
Cope & drag	Separate patterns for cope and drag mounted on separate plates; for heavy, big castings.
Sweep	A thin board rotated about a spindle to form large symmetrical moulds (bells, kettles) - saves pattern cost.
Skeleton	Wooden frame/skeleton (ribs); sand is filled & struck off - very large, few-off castings.
Shell (shell mould)	Metallic pattern, heated, for shell moulding process.

Materials used for patterns

Material	Merits	Demerits / Use
Wood (teak, pine, deodar, mahogany)	Cheap, light, easy to shape & repair	Absorbs moisture, wears quickly; for 1-100 castings
Metals (Al, brass, cast iron, steel)	Durable, accurate, smooth, high strength	Costly, heavy; mass production (Al most common)
Plastics (epoxy, polyurethane)	Light, no moisture, smooth finish	Less strength; medium run
Wax	Melts out easily, fine detail	Used in investment casting
Plaster of Paris	Intricate shapes, easy to cast	Weak, brittle; small castings
Rubber / Polystyrene	Flexible; expanded polystyrene vaporises	Used in full-mould / lost-foam

Pattern allowances

Dimensions of a pattern are changed from the casting dimensions by the following allowances:

Pattern Allowances (very important - 5 M)

- Shrinkage (contraction) allowance: metal shrinks on cooling from solidification temperature to room temp, so pattern is made larger. Positive allowance. Use a shrink rule (longer than a standard scale). Typical: grey C.I. 1 % (10 mm/m); steel 2 % (20 mm/m); aluminium 1.5 %; brass 1.6 %.
- Machining (finish) allowance: extra metal on surfaces to be machined (about 2-6 mm depending on size, metal & finish required).
- Draft (taper) allowance: slight taper (1-3 degree, or 10-20 mm/m on external surfaces; more on internal) on vertical faces so that the pattern can be withdrawn without damaging the mould.
- Shake (rapping) allowance: pattern is rapped to loosen it in the mould, which enlarges the cavity - so pattern is made slightly smaller (negative allowance, ~0.5-1 mm).
- Distortion allowance: for shapes like U, V, thin long casting which distort due to uneven cooling - pattern is given an opposite curvature.
- Core-print allowance: extra projection on pattern to provide seat for the core.

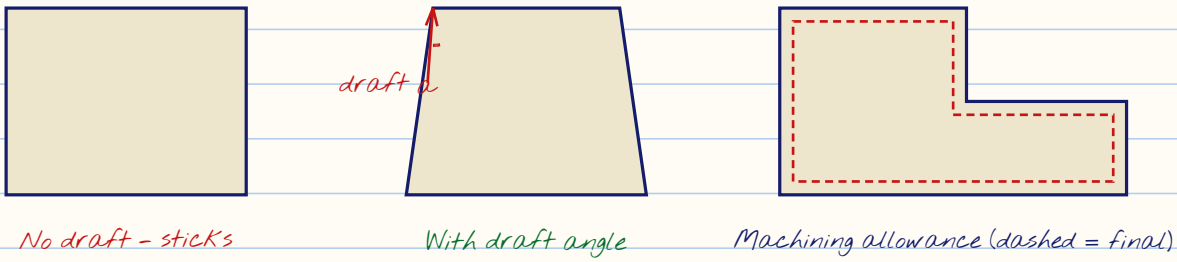


Fig 2.2 Draft & machining allowance

Pattern construction & colour code

- Wood patterns are built from well-seasoned wood joined by glue, screws, dowels; edges rounded with fillets (to avoid sharp corners, tearing & hot spots); surface varnished/shellac.
- Large patterns are built-up or skeleton; thick sections are hollowed to cut weight.
- Colour code:

Colour	Meaning
Red / Orange	Surfaces to be machined
Black	Un-machined (cast) surfaces
Yellow	Core prints / seats for loose pieces
Red stripes on yellow	Seats for loose pieces (stop-offs)
Black stripes on yellow	Stop-offs (spaces to be filled)

3. Molding (Moulding)

Moulding is the process of making a mould cavity by packing moulding sand around the pattern in a moulding box (flask) and withdrawing the pattern.

Moulding sand - composition & properties

- Composition: silica sand (SiO_2) 75-85 %, clay binder (bentonite / fire clay 10-20 %), moisture (2-8 %) and additives (coal dust, wood flour, dextrin, molasses).
- Types: green sand (moist, most common), dry sand (baked), loam, facing sand, backing sand, parting sand (dry silica to avoid sticking), core sand.

Properties of moulding sand

- Permeability: allows gases & steam to escape. - Refractoriness: withstand high temp without fusing.
- Cohesiveness: sand grains stick together. - Adhesiveness: sticks to the box.

- Flowability: fills pattern details. - Collapsibility: yields as casting shrinks (avoids hot tears).
- Green strength / Dry strength, good thermal stability, chemical inertness, reusability.

Steps in making a green-sand mould

- Place pattern (drag half) on moulding board in the drag box; sprinkle parting sand.
- Fill facing sand then backing sand; ram it uniformly; strike off excess; turn the drag over.
- Place cope half of pattern, sprue pin & riser pin; fill, ram, vent with vent wire (for gases).
- Remove sprue/riser pins, cut pouring basin; separate cope & drag; draw out the pattern gently (using draw spikes, rapping).
- Cut runner & gates; place cores; close the mould, clamp / weigh down the cope - ready for pouring.

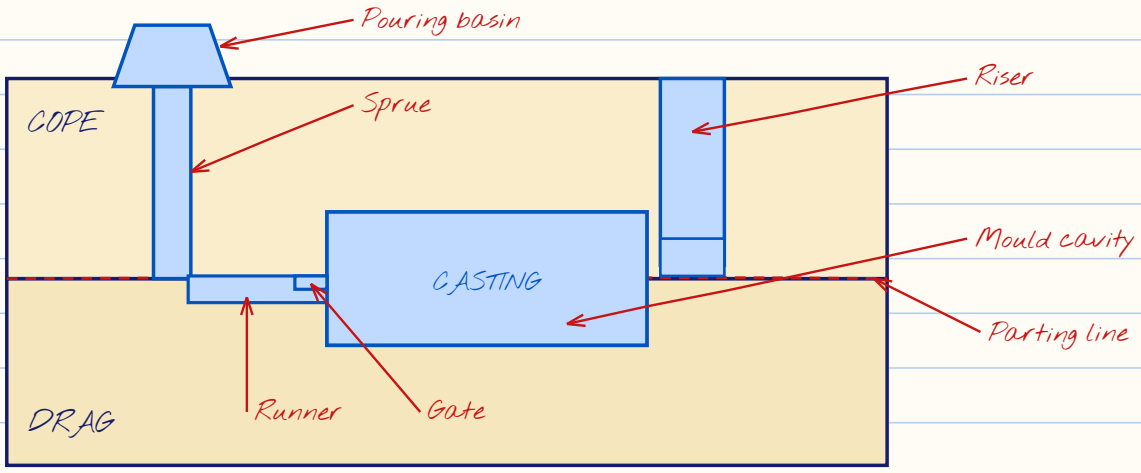


Fig 3.1 Sand mould with gating system

Types of moulding / moulds

- Bench / floor / pit moulding: small jobs on bench; medium on floor; very large castings in pits (pit itself acts as drag).
- Machine moulding: squeeze, jolt, jolt-squeeze, sand-slinger - for mass production, uniform ramming.

- Green sand mould: damp, cheapest, good collapsibility but gives lower strength & dimensional accuracy.
- Dry sand mould: baked at 150-300 degree C - stronger, more accurate, for large castings; costly, slow.
- Loam, CO_2, shell & no-bake moulds: loam for huge symmetrical parts; CO_2 process hardens sand (sodium silicate) quickly; no-bake uses resin binders.

4. Cores & Core Making

A core is a pre-formed body of baked sand placed in the mould cavity to produce internal cavities / holes / undercuts in the casting. Core sand contains silica sand + binders (linseed oil, core oil, cereal, resin, sodium silicate) - baked at 200-250 degree C to give high strength.

Requirements: adequate strength, permeability, refractoriness, collapsibility, smooth surface, low gas emission.

Types of cores

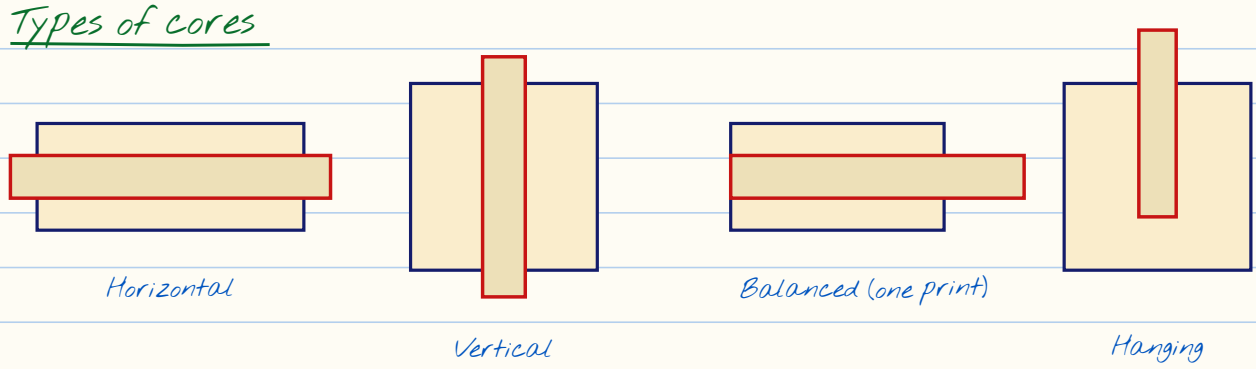


Fig 4.1 Some types of cores (red = core)

Core	Remarks
Horizontal	Placed horizontally; supported at both ends by core prints in the parting plane.
Vertical	Axis vertical; core prints on cope and drag - for holes in bushes, gear-blanks.
Balanced	Supported at only one end; its weight is balanced by a longer print.
Hanging (cover)	Hangs from cope; no support in drag. Cover core lies over the mould.
Drop / Ram-up	Drop core is placed from cope, below parting line. Ram-up core is rammed in with the pattern.
Kiss core	Core held in place by pressure between cope and drag, no core print.
Wing / Green-sand	Wing core for projections; green sand core formed by the pattern itself.

- Core prints: extra projections on pattern which form seats in mould to support the core.
- Chaplets: small metal supports (same material as casting) fixed between mould & core to prevent core sagging/floating; they fuse into the casting.
- Core making: sand mixed with binder, shaped in a core box by hand/ blowing machines (core blower / shooter), baked in core oven, finished & coated (blacking/ graphite), vented using wax wire.

5. Gating System, Risers & Casting Design

Gating system - principles

The gating system is the network of channels (pouring basin, sprue, runner, gates) that leads molten metal into the cavity. A good gating system should:

- Fill the cavity quickly & smoothly without turbulence, thereby avoiding oxidation, gas pick-up, erosion of sand (cuts & washes).
- Establish directional solidification (thin section/ far-end first, riser last).
- Trap slag, dross & inclusions (using skim bob, strainer core, choke, slag traps).
- Avoid air aspiration - the sprue should be tapered and have a well at the bottom.
- Maintain minimum metal consumption (high casting yield).

Elements of gating system

- Pouring cup / basin: receives metal from ladle, reduces turbulence & helps in skimming slag.
- Sprue: vertical tapered channel; velocity $v = \sqrt{2gh}$; continuity $A1\,v1 = A2\,v2$ so $A2/A1 = \sqrt{(h1/h2)}$ (tapered to avoid aspiration).
- Sprue well: enlarged base to absorb energy of falling metal.
- Runner: horizontal channel distributing metal to gates. - Gates (ingates): connect runner with cavity; control rate of filling, shape for easy removal.
- Riser: reservoir of molten metal feeding shrinkage.

Types of gating systems

- Top gate: metal enters from the top - promotes directional solidification but causes splashing/erosion (for short, simple castings).
- Bottom gate: smooth, quiet filling, little turbulence; but unfavourable temperature gradient.

- Parting-line (side) gate: most common compromise - at the parting plane; Step gate: multiple gates at different heights for tall castings.
- Pressurised vs un-pressurised: pressurised (gate area < runner area; ratio sprue:runner:gate = 1 : 0.75 : 0.5) - fast filling, for ferrous; un-pressurised (1 : 3 : 3 or 1:2:1) - less turbulence for Al/Mg alloys.
- Filling time $t = W / (\rho A_c v)$. Choke area $A_c = W / (\rho \cdot C \sqrt{2 g H})$.

Risers

A riser (feeder / feed head) is a reservoir of molten metal connected to the casting to compensate for liquid and solidification shrinkage. It must solidify after the casting. Functions: feeds shrinkage, lets out air/gases, indicates the mould is full, helps directional solidification.

- Types: top riser, side riser, blind riser (inside cope; has a atmospheric core - for better feeding), open / closed riser. Shape: cylindrical is best (min. surface area for volume - slow cooling).
- Chvorinov's rule: Solidification time $\{r|t = B (V/A)^2\}$ where V = volume, A = cooling surface area, B = mould constant. Riser must have larger V/A than the casting: $(V/A)_{riser} >= 1.2 (V/A)_{casting}$.
- Caine's method: freezing ratio $X = (A/V)_{casting} / (A/V)_{riser} = a / (Y - b) + c$, where Y = riser volume / casting volume.
- Chills: metallic inserts (internal / external) placed in the mould to accelerate cooling at thick sections and promote directional solidification; Padding: taper added to thin sections to help feed. Insulating / exothermic sleeves keep riser hot longer.

Casting design considerations

- Provide uniform section thickness; avoid abrupt changes (use gradual tapering, 1:4 ratio) - prevents hot spots & shrinkage cavities.
- Use fillets & rounded corners (no sharp edges - cracks & hot tears); avoid large flat surfaces (warpage - add ribs).
- Sections should join at proper angles; avoid intersection of many ribs at one point (hot spot) - stagger them.
- Choose parting line to be flat, simple & at the largest cross-section; provide draft.
- Design to minimise cores, avoid undercuts; provide machining allowance &

datum faces for machining.

- Locate thick parts near risers, to ensure directional solidification; avoid isolated heavy bosses.

6. Melting Methods & Furnaces

Furnace selection depends on type of metal, quantity, melting temperature, quality, cost & fuel. Common melting furnaces:

Furnace	Used for	Features
Cupola	Grey cast iron	Vertical shaft, continuous melting, cheap, coke fuel
Crucible (pit / tilting)	Non-ferrous (Al, Cu alloys), small	Oil / gas / coke fired; small batches; easy alloying
Electric arc	Steel, alloy steel, C.I.	Fast, high temp (up to 1800 C), good quality control, big capacity
Induction (coreless / channel)	Steel, C.I., non-ferrous	Clean, stirring by eddy currents, accurate composition
Reverberatory (air)	Malleable iron, Al scrap, large lots	Flame deflected from roof; large capacity, low cost
Pot / open-hearth	Steel (older)	Regenerative heating; slow but large

Cupola furnace

A vertical cylindrical furnace of steel shell lined with refractory (fire brick). Charge (pig iron + scrap + coke + limestone flux) is fed from the charging door at top in alternate layers; air blast through tuyeres near bottom burns coke and melts the iron which collects in the well and is tapped from the spout; slag is removed through the slag hole.

- Fuel : metallurgical coke (8–16 % of metal); flux limestone ($CaCO_3$) 2–3 % removes ash as slag.
- Capacity expressed in tonnes/hour of melt; blast rate ~ 30 m^3/min per m^2 of cross-section.
- Advantages: simple, continuous, economical, high melting rate. Limits: little control of composition (picks up S, loses Si/Mn), only cast iron.

Other furnaces

- Crucible: metal in clay-graphite / SiC pot heated externally; no contact with

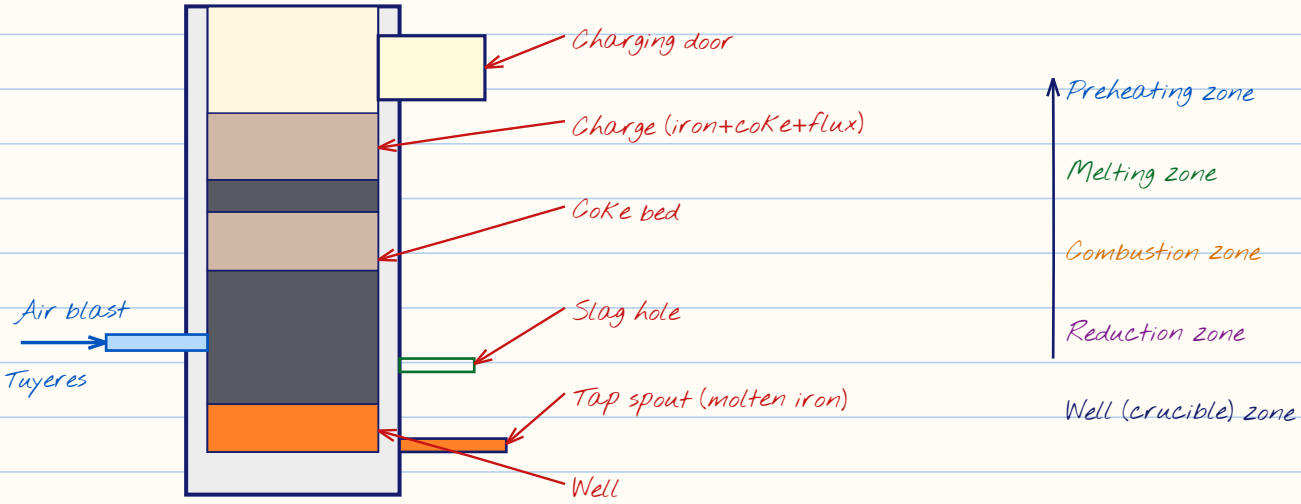


Fig 6.1 Cupola furnace

fuel => clean melt; tilting type pours directly.

- Electric arc (direct): three graphite electrodes strike arc with charge; temp up to 3000 C at arc; basic lining for steel (removes P,S).
- Induction: AC in a coil induces eddy currents in the charge placed in refractory crucible; stirring ensures uniform composition; low-frequency (channel) & high-frequency (coreless).

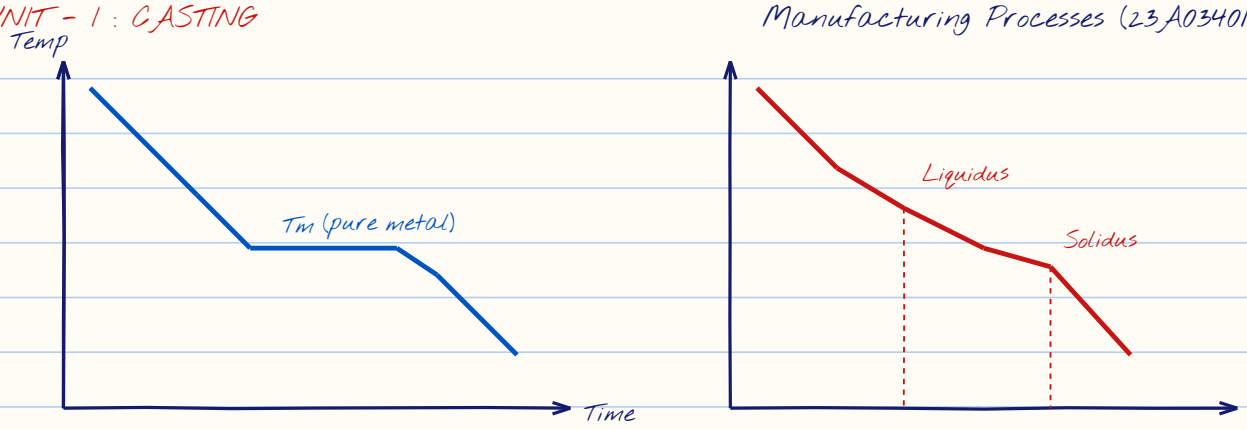
7. Solidification of Castings

- Pure metal: solidifies at a single temperature (constant); skin forms first at mould wall; columnar growth towards centre; shrinks and forms a pipe.
- Alloy: solidifies over a freezing range (liquidus - solidus) - mushy zone with dendrites; larger range => more porosity; skin may not form.
- Nucleation & growth: grain structure - chill zone (fine equi-axed at wall), columnar zone, central equi-axed zone. Fine grains give better strength: use inoculants / grain refiners (Ti, B in Al).
- Shrinkage: three stages - liquid contraction, solidification contraction (cavities), solid contraction (pattern allowance).
- Directional solidification: far end -> riser; progressive, using chills, risers, padding.

Chvorinov's rule (remember!)

Solidification time $t_s = B (V / A)^2$ (V = volume of casting, A = surface area cooling, B = mould constant)

- Riser should freeze later => riser V/A larger than casting V/A.
- A sphere has the highest V/A => longest time; thin plate shortest.



Pure metal - cooling curve

Alloy - freezing range

Fig 7.1 Cooling curves

8. Casting Defects - Causes & Remedies

Defect	Causes	Remedies
Blow holes / gas porosity	Trapped gases (moisture, high pouring temp, low permeability, poor venting)	Dry mould, increase permeability, proper venting, degas metal, lower pouring temp
Shrinkage cavity	Insufficient feeding, poor riser, non-uniform section	Proper riser size/location, chills, directional solidification, uniform section
Misrun	Metal freezes before filling cavity - low pouring temp, thin section, slow pouring	Raise pouring temp, increase fluidity & section thickness, improve gating
Cold shut	Two streams of metal meet but do not fuse (low temp / slow pouring)	Higher pouring temp, faster pouring, change gate position
Hot tear (crack)	Restricted contraction in a hot, weak casting; sharp corners	Collapsible sand/core, fillets, proper cooling, uniform sections
Cuts & washes	Erosion of sand by high-velocity metal	Reduce velocity, use strong sand, proper gating (well)
Inclusions / sand holes / dross	Loose sand, slag, dross entering cavity	Clean mould, skim, strainer/filters, good gating, proper ramming
Scab / swell / drop	Sand surface layer lifts due to hot metal; weak mould wall; soft ramming	Proper ramming, good permeability, correct moisture, strong sand
Mould/core shift	Misalignment of cope-drag, core displaced	Proper alignment (pins), clamps, chaplets, secure core
Fins / flash	Metal leaking along parting line (cope not weighted)	Clamp/ weight properly, good fit of box halves

Warpage / distortion	Uneven cooling, large flat areas	Ribs, proper pattern curvature, uniform cooling
Metal penetration / fusion	Very high temp, coarse sand, poor sand strength	Fine sand, mould wash/ coating (blacking), lower temperature

9. Special Casting Processes

9.1 Centrifugal casting

Molten metal is poured into a rotating mould; centrifugal force throws metal against the mould wall and it solidifies under high pressure giving dense, impurity-free castings (impurities, being lighter, collect at the bore & are machined away).

- True centrifugal: axis of rotation = axis of casting; hollow cylinders (pipes, liners, bushes) without cores. Horizontal (de Lavaud) / vertical axis. Speed N: G-factor (rw^2/g) 60–80 (horizontal).
- Semi-centrifugal: solid-type castings (wheels, pulleys, gear blanks) with a central core/riser; mould spins about vertical axis.
- Centrifuging: several small cavities arranged around a central sprue; metal forced into them by rotation – small, intricate parts (jewellery).
- Pros: no cores, dense fine grain, good mechanical properties, no gating loss. Cons: only symmetrical shapes; inner diameter inaccurate; segregation of alloys. Uses: pipes, cylinder liners, bushings, rolls.

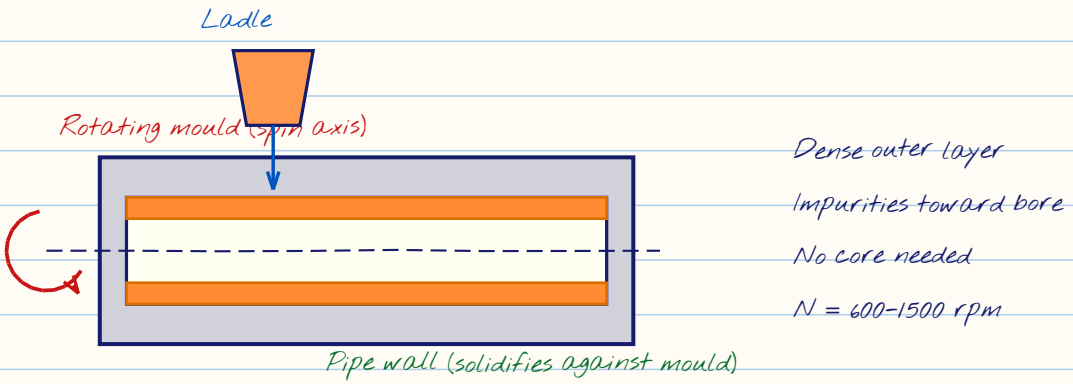


Fig 9.1 True centrifugal casting (horizontal)

9.2 Die casting (pressure die casting)

Molten metal is forced under high pressure (7-700 MPa) into a steel die (permanent mould) and held until it solidifies. Used mainly for low melting non-ferrous alloys - Zn, Al, Mg, Cu, Pb, Sn.

Hot-chamber	Cold-chamber
Injection cylinder (gooseneck) immersed in molten metal in the furnace; plunger forces metal into die.	Metal is ladled into a separate shot chamber, then plunger injects at high pressure.
For low-melting alloys (Zn, Sn, Pb); fast cycle (up to 500 shots/h); pressure 7-35 MPa.	For Al, Mg, Cu alloys (they attack the plunger/pot); pressures up to 150 MPa +; slower cycle.

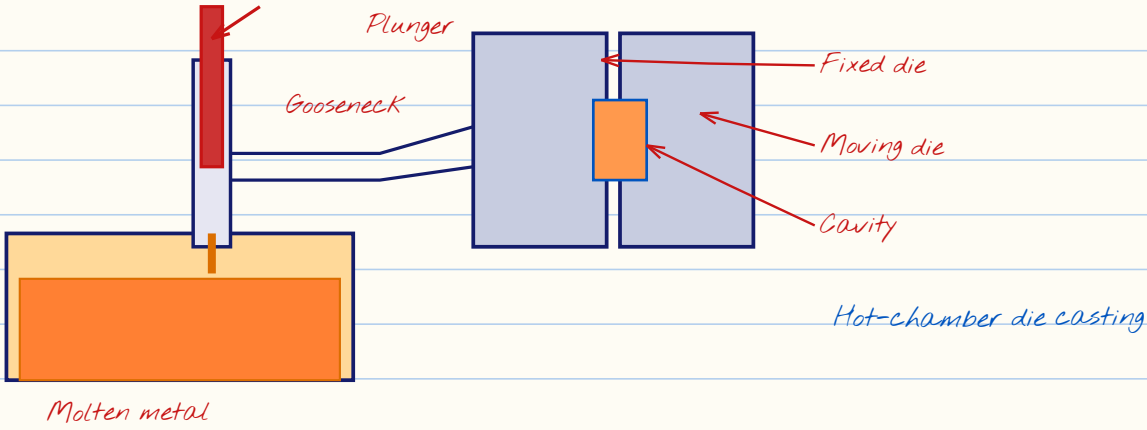


Fig 9.2 Hot-chamber die casting machine

- Advantages: high production rate, excellent dimensional accuracy & surface finish, thin walls, little machining, long die life.
- Limitations: high die/machine cost (economical only for large volume), size & metals limited, air entrapment porosity. Uses: carburettor, gear-box cases, toys, hardware, automobile & appliance parts.

9.3 Investment casting (lost-wax process)

Used for intricate, precise parts - turbine blades, jewellery, dental parts, surgical tools. Pattern is of wax which is melted out ("lost") from a ceramic shell.

- Merits: excellent accuracy (+-0.1 mm) & finish, extremely complex shapes, any alloy (even high-melting, superalloys), no parting line. Demerits: costly, slow, size limited (< 5-10 Kg usually), many steps.

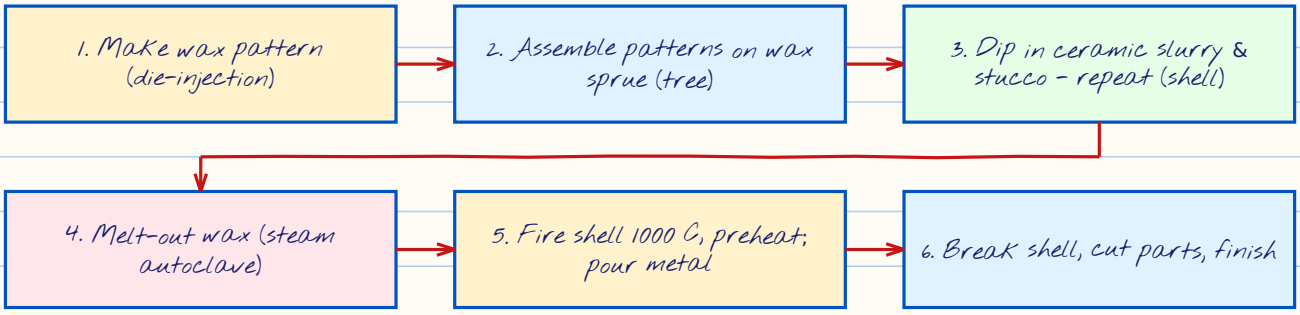


Fig 9.3 Steps in investment casting

9.4 Shell moulding (Croning process)

A thin shell (~5-10 mm) of sand + thermosetting resin (phenol-formaldehyde) is made over a heated metal pattern (200-250 C); shell halves are cured, joined and backed with sand/shot, then metal is poured.

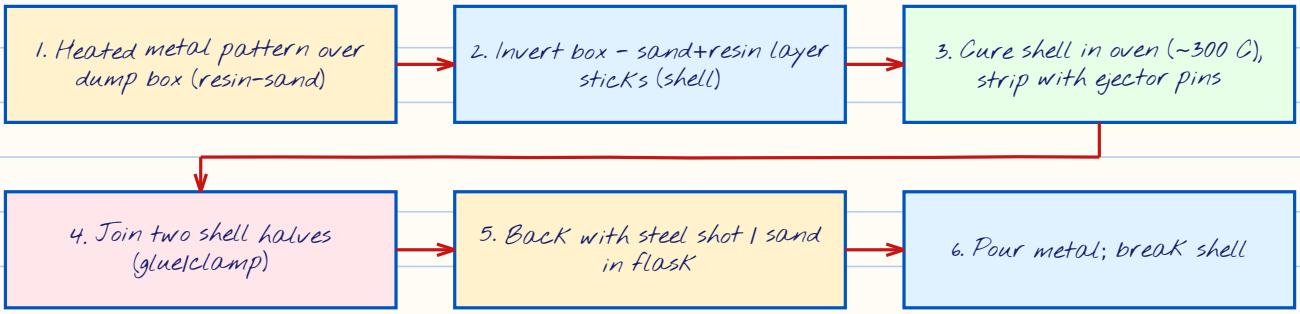


Fig 9.4 Shell moulding steps

- Advantages: good surface finish & accuracy (+-0.25 mm), thin sections, high production, low sand consumption, easy automation, no ramming skill. Limits: costly metal pattern & resin, size limited, fumes. Uses: gear housings, cylinder heads, connecting rods, camshafts, small automotive parts.

Important exam questions - Unit I

- Explain the steps involved in making a casting. List advantages & applications. (5 M)
- What is a pattern? Types of patterns & materials used. Explain allowances with examples. (10 M)
- Describe the principles of gating; design of sprue; types of gating systems. (10 M)
- Chvorinov rule & riser design; role of chills. (5 M)
- Explain the cupola furnace with a neat sketch. (10 M)
- Casting defects: causes & remedies (5-10 M).
- Centrifugal / die / investment / shell casting - principle, steps, merits & applications (10 M each).